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Close-Range Rendezvous with a Moving Target Spacecraft using Udwadia-Kalaba Equation*

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Abstract— **This paper presents an analytical dynamics based formulation for close-range planar rendezvous of two chaser spacecraft onto an uncontrolled target spacecraft. The control requirements on the chaser system is formulated based on displacement constraints with respect to target. Exact control forces are then generated based on the acceleration constraint equation, which is derived from the displacement constraint, and substituted into the Udwadia-Kalaba equation. As the major contribution, a new formulation is developed for both a single and dual-chaser close-range rendezvous. The simulation results highlight the simultaneous angular velocity synchronization and stand-off distance maintenance with respect to the target.**

I. Introduction

In recent years, there has been a rising concern about the growth of space debris in lower-earth orbit. The insightful prediction by Kessler et al. in 1978 about the development of a debris belt surrounding the earth and consequent exponential growth of new debris from collision and explosions [1] is gradually becoming a reality. Recently, the European Space Agency reported that the number of objects in lower earth orbit has reached a critical concentration that calls for remediation measures through active debris removal (ADR)†. In the ADR technique, a chaser spacecraft is used to rendezvous and capture a piece of tumbling debris. Removing such objects reduce the collision probability and allow orbits to remain usable for future space missions.

Typically, an ADR mission involves a five-phase process of chaser launch, far-range rendezvous around the target followed by a close-range rendezvous, debris capture and removal [2]. For the post-capture detumbling, use of tethered spacecraft systems (TSS) is being considered as a promising option due to the tether flexibility, low weight and ability to handle varying debris size [3], [4], [5]. In all of these methods, however, it was noted that tethers are unable to exert any force under compression. Hence, a complete inertial stabilization of debris was impossible with a single chaser and target system. A potential solution is to use multiple chasers to pull on the target so as to maintain the tension. In order to enable the practical implementation of the above techniques, the focus of this paper is to propose a new methodology to accomplish the close-range rendezvous phase using two chasers based on recent advancements in analytical dynamics. The proposed method achieves the desired alignment while maintaining a specified stand-off distance with respect to the target spacecraft. This would allow for the capture and inertial detumbling maneuvers to be ultimately performed.

The requirement to maintain a constant separation while tracking targets in surveillance operations have motivated the study of stand-off tracking in the guidance of unmanned aerial vehicles[6], [7]. The methods in this domain attempt to simultaneously control the position and orientation of the vehicle approach with respect to a point mass target. While a point mass assumption is useful in a wide range of conceptually similar space applications like satellite formation maintenance [8], orbit transfer studies [9] and far-range rendezvous [10], such an assumption is not sufficient for operations like docking or pose tracking when closer to the target. Due to the need for precise positioning and alignment with respect to the target before docking, the close-range rendezvous phase has frequently been studied in on-orbit servicing applications [11]. While approaching near a target, the requirement for accurate control of its closing trajectory becomes mandatory. The chaser needs to be aligned and it's linear motion controlled so as to capture the debris with a zero relative angular motion at the specified stand-off distance. The resulting linear and angular motions show low-frequency coupling and hence, require a multivariable strategy. In traditional approaches, this cross-coupling term is input as a feed-forward term following which a classical PID type of control is enforced. H_∞ design approaches are also frequently used to have an efficient system. In the terminal approach phase, however, the high-frequency couplings call for multi-variable design approaches. An experimental study of close-range rendezvous and docking for a cooperative target was presented by Romano et al. [12]. Ma et al. [13] presented a feed-forward control to achieve zero relative motion to capture an arbitrarily tumbling target in close-range rendezvous. Using the Pontryagins Maximum Principle for combined time and fuel-optimization, a reasonable trajectory was achieved for the closing phase of rendezvous. The cascading of dual objective iterations, however, drove up the computational cost in this approach. Wilde et al. [14] experimentally demonstrated an inverse dynamics based trajectory optimization for planar docking of a chaser onto an uncontrolled spinning target. Ventura et al. [15] presented a spatial inverse-dynamics based approach for docking onto a tumbling object where the inversion is done in a virtual domain rather than a time domain for the iterative generation

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of optimal trajectories. Using the current state estimates, the approach trajectory is calculated and the control is updated. Zagaris and Romano [16] presented the reachability analysis for planar docking onto a spinning target. In the domain of spacecraft formation flying, Yan et al. [17] presented pulse-based linear feedback control on a discretized model of the Clohessy-Whiltshire equations that demonstrated stable convergence characteristics. Naasz et al [18] introduced an H_2/H_∞ control strategy for formation maintenance in the presence of orbital perturbations based on the Clohessy-Whiltshire equations. Lyapunov arguments were used to show a global asymptotic stability.

This work presents a new close-range rendezvous method that uses recent advancements in analytical dynamics to simultaneously control the linear and rotational motion to precisely position and align multiple chasers with respect to a target. Specifically, the Udwadia-Kalaba equation (UKE) [19], [20], [21] is exploited to generate exact force commands that obeys the set performance characteristics while achieving the desired coupled motion in real-time; thereby enabling a dual-chaser rendezvous maneuver to a spinning target. Furthermore, simulations are carried out to demonstrate the approach.

The remainder of the paper is arranged as follows: Section II describes the rendezvous problem. Proposed Udwadia-Kalaba equation based rendezvous formulation is presented in Sec.IV. Simulation studies are discussed in Sec. V following which the concluding remarks are drawn in Sec. VI.

II. Problem Formulation

It assumed that the position and velocity information of debris is known with sufficiently high accuracy. Acceleration of the debris could be calculated by incorporating on-orbit effects that depend on position data, if required. Recent advancements in computer vision could be combined with the state data from multiple chaser cameras to interpret the pose of tumbling target. The relative velocities between the chasers and debris are assumed to be small so that as to attain the capture without control saturation. The external forces or torques are ignored on account of a larger magnitude of the thrust force or torque. The docking maneuver is assumed to be fast as compared to the orbital period of the target so that the relative dynamics can be ignored over a shorter time period. The double integrator dynamics is considered for position and orientation dynamics.

Consider a plane $\mathcal{S} \subseteq \mathbb{R}^2$ on which the two chasers and the target are located, as illustrated in Fig. 1. The bodies are located at (x_k, y_k) at an orientation θ_k where the subscript $k = c1, c2$ or t represent Chaser 1, Chaser 2, or target, respectively. The corresponding linear velocity and acceleration vectors are $(\dot{x}_k, \dot{y}_k)$ and $(\ddot{x}_k, \ddot{y}_k)$. The angular velocity and angular accelerations are represented by $\dot{\theta}_k$ and $\ddot{\theta}_k$. The bodies have a mass m_k and an out-of-plane inertia of J_k. As the net external force acting in close-range rendezvous is small, the target velocity is assumed to be fixed. In the first part of the problem, it is desired to get Chaser 1 to synchronize with the target angular velocity

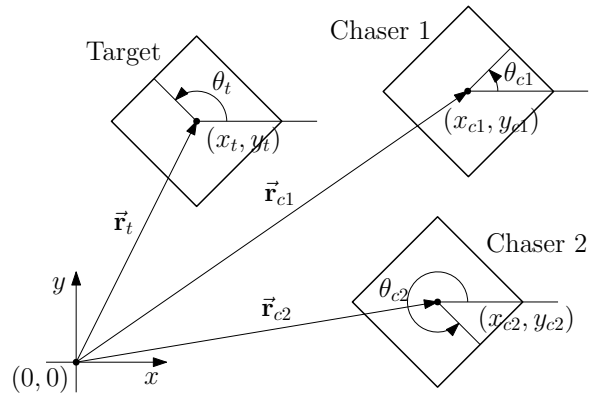

Fig. 1: Problem formulation

while maintaining a constant stand-off distance L_0 at a given relative orientation with respect to the target. In second part, the chasers are desired to synchronize their angular velocities and rendezvous with the target while maintaining a diametrically opposite location with respect to the target center of mass. Prospective formulation is desired to provide exact real-time force commands to achieve the desired trajectory.

III. Udwadia-Kalaba Framework

This section reviews the Udwadia-Kalaba framework [19] and details the fundamental components of the control methodology. Suppose a given generalized force $\mathbf{F}_u$ is acting upon a system with generalised mass $\mathbf{M}$, the unconstrained generalized acceleration is

$$\ddot{\mathbf{q}}_u = \mathbf{M}^{-1}\mathbf{F}_u \tag{1}$$

On applying a set of m acceleration constraints as given by

$$\mathbf{A}_m\ddot{\mathbf{q}} = \mathbf{b}_m \tag{2}$$

The constraint acceleration is obtained using the Udwadia-Kalaba equation as

$$\ddot{\mathbf{q}}_c = \mathbf{M}^{-1/2}(\mathbf{A}_m\mathbf{M}^{-1/2})^+(\mathbf{b}_m - \mathbf{A}_m\mathbf{M}^{-1}\mathbf{F}_u) \tag{3}$$

where the superscript $^+$ indicate pseudo-inversion of non-square matrix. Equation (3) is referred to as the Udwadia-Kalaba equation. The effective motion is then described by

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_u + \mathbf{M}^{-1/2}(\mathbf{A}_m\mathbf{M}^{-1/2})^+(\mathbf{b}_m - \mathbf{A}_m\ddot{\mathbf{q}}_u) \tag{4}$$

Alternately, Eq. (4) could also be written as [20]

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_u + \mathbf{M}^{-1}\mathbf{A}_m^\intercal(\mathbf{A}_m\mathbf{M}^{-1}\mathbf{A}_m^\intercal)^+(\mathbf{b}_m - \mathbf{A}_m\ddot{\mathbf{q}}_u) \tag{5}$$

The idea that control action is essentially a constraint on the states of a system was exploited by Koganti et al. [21] to generate desired force commands for a generic tumbling multi-body system. Udwadia-Kalaba equation (UKE) was applied on the control constraint to generate exact real-time commands to achieve the desired trajectory with the set performance characteristics. Suppose the control objective is to get one of the actual states, q_k to the desired state $q_{k,d}$. Then, the control constraint i is defined as

$$\phi_i := q_k - q_{k,d} \tag{6}$$

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Control action is then desired to force $\phi_i \to 0$. The first derivative of Eq. (6) is given by

$$\dot{\phi}_i := \dot{q}_k - \dot{q}_{k,d} \tag{7}$$

On differentiating Eq. (7),

$$\ddot{\phi}_i := \ddot{q}_k - \ddot{q}_{k,d} \tag{8}$$

Using Eq. (6)-(8), a second order error dynamics equation could be built as

$$\ddot{\phi}_i + \alpha_i \dot{\phi}_i + \gamma_i \phi_i = 0 \tag{9}$$

where $\alpha_i > 0$ and $\gamma_i > 0$ are constants. Note that, due to the non-negative values for α_i and γ_i, the second order dynamical system results in $\phi_i \to 0$. The value of α_i and γ_i could be chosen to achieve the desired performance characteristics. Although theoretically, any non-negative choice of the control gains would work, in reality, the choice is to be guided by the control saturation limits of the thrusters. Substituting Eq. (6)-(8) into Eq. (9) and rearranging,

$$\ddot{q}_k = \ddot{q}_{k,d} - \alpha_i(\dot{q}_k - \dot{q}_{k,d}) - \gamma_i(\dot{q}_k - \dot{q}_{k,d}) \tag{10}$$

In the standard form, the ith control constraint could be written as

$$\mathbf{A}_{c,i}\ddot{\mathbf{q}} = b_{c,i} \tag{11}$$

where $\mathbf{A}_{c,i}$ is a $1 \times n$ matrix with the element multiplying with the q_k^{th} term as 1 and the rest 0, $\mathbf{q}$ is the generalized coordinate, and

$$b_{c,i} = \ddot{q}_{k,d} - \alpha_i(\dot{q}_k - \dot{q}_{k,d}) - \gamma_i(\dot{q}_k - \dot{q}_{k,d}) \tag{12}$$

For a set of m constraints, Eq. (11) is composed into a single control constraint matrix equation as

$$\mathbf{A}_c\ddot{\mathbf{q}} = \mathbf{b}_c \tag{13}$$

where $\mathbf{A}_c = \begin{bmatrix}\mathbf{A}_{c,1}; & \mathbf{A}_{c,2}; & \dots & \mathbf{A}_{c,m}\end{bmatrix}_{m\times n}$ and $\mathbf{b}_c = \begin{bmatrix}b_{c,1} & b_{c,2} & \dots & b_{c,m}\end{bmatrix}^{\intercal}_{m\times 1}$. Using UKE, the exact control force to achieve the desired state at the specified performance is

$$\mathbf{F}_c = \mathbf{A}_c^{\intercal}(\mathbf{A}_c\mathbf{M}^{-1}\mathbf{A}_c^{\intercal})^{+}(\mathbf{b}_c - \mathbf{A}_c\mathbf{M}^{-1}\mathbf{F}_u) \tag{14}$$

IV. Udwadia-Kalaba-Based Guidance Methodology

To begin, the desired final rendezvous configuration is chosen which forms the basis for building the constraint function. Following this, the first and second derivatives of the constraint function are composed into the standard constraint form which can be used the generate the control forces based on UKE.

For Chaser 1, the final configuration of rendezvous is as shown in Fig. 2. The corresponding states are written as

$$\begin{bmatrix} x_{c1} \\ y_{c1} \\ \theta_{c1} \end{bmatrix} = \begin{bmatrix} x_t + L_0\cos\theta_t \\ y_t + L_0\sin\theta_t \\ \theta_t + \pi \end{bmatrix} \tag{15}$$

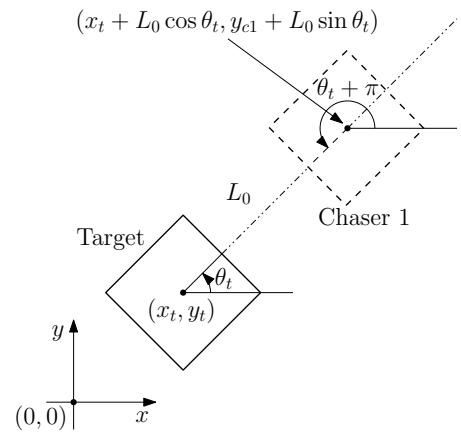

Fig. 2: Desired rendezvous configuration for single chaser.

Three constraint equations could then be defined for Chaser 1 as

$$\begin{aligned} \phi_1 &= x_{c1} - x_t - L_0\cos\theta_t && (16)\\ \phi_2 &= y_{c1} - y_t - L_0\sin\theta_t && (17)\\ \phi_3 &= \theta_{c1} - \theta_t - \pi && (18) \end{aligned}$$

Differentiating (16), (17) and (18) with respect to time,

$$\begin{aligned} \dot{\phi}_1 &= \dot{x}_{c1} - \dot{x}_t + L_0\sin\theta_t\dot{\theta}_t && (19)\\ \dot{\phi}_2 &= \dot{y}_{c1} - \dot{y}_t - L_0\cos\theta_t\dot{\theta}_t && (20)\\ \dot{\phi}_3 &= \dot{\theta}_{c1} - \dot{\theta}_t && (21) \end{aligned}$$

Double differentiating Eqs. (16), (17) and (18) with respect to time gives

$$\begin{aligned} \ddot{\phi}_1 &= \ddot{x}_{c1} - \ddot{x}_t + L_0\sin\theta_t\ddot{\theta}_t + L_0\cos\theta_t\dot{\theta}_t^2 && (22)\\ \ddot{\phi}_2 &= \ddot{y}_{c1} - \ddot{y}_t - L_0\cos\theta_t\ddot{\theta}_t + L_0\sin\theta_t\dot{\theta}_t^2 && (23)\\ \ddot{\phi}_3 &= \ddot{\theta}_{c1} - \ddot{\theta}_t && (24) \end{aligned}$$

Now, considering the second chaser, the desired configuration is set as shown in Fig. 3. This arrangement is meant to ensure that tethers, when deployed to catch the target, are always in tension during the post-capture phase. The final configuration for Chaser 2 would then be written as

$$\begin{bmatrix} x_{c2} \\ y_{c2} \\ \theta_{c2} \end{bmatrix} = \begin{bmatrix} x_t - L_0\cos\theta_t \\ y_t - L_0\sin\theta_t \\ \theta_t \end{bmatrix} \tag{25}$$

The corresponding three constraint equations be composed as

$$\begin{aligned} \phi_4 &= x_{c2} - x_t + L_0\cos\theta_t && (26)\\ \phi_5 &= y_{c2} - y_t + L_0\sin\theta_t && (27)\\ \phi_6 &= \theta_{c2} - \theta_t && (28) \end{aligned}$$

The first derivatives are obtained as

$$\begin{aligned} \dot{\phi}_4 &= \dot{x}_{c2} - \dot{x}_t - L_0\sin\theta_t\dot{\theta}_t && (29)\\ \dot{\phi}_5 &= \dot{y}_{c2} - \dot{y}_t + L_0\cos\theta_t\dot{\theta}_t && (30)\\ \dot{\phi}_6 &= \dot{\theta}_{c2} - \dot{\theta}_t && (31) \end{aligned}$$

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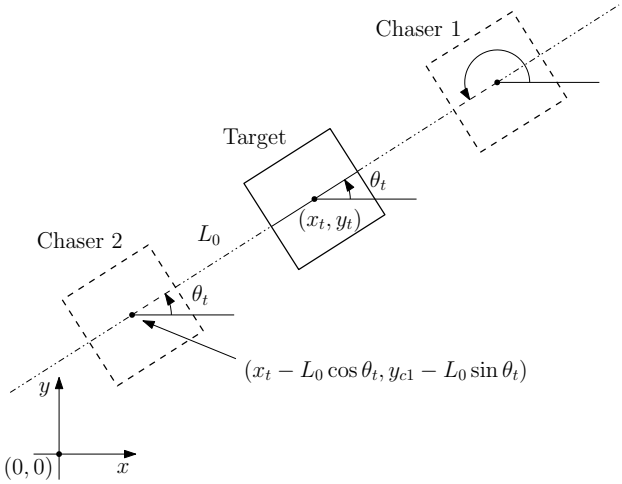

Fig. 3: Desired rendezvous configuration for dual-chaser system.

Second derivative of the constraint functions give

$$\ddot{\phi}_4 = \ddot{x}_{c2} - \ddot{x}_t - L_0 \sin\theta_t \ddot{\theta}_t - L_0 \cos\theta_t \dot{\theta}_t^2 \quad (32)$$
$$\ddot{\phi}_5 = \ddot{y}_{c2} - \ddot{y}_t + L_0 \cos\theta_t \ddot{\theta}_t - L_0 \sin\theta_t \dot{\theta}_t^2 \quad (33)$$
$$\ddot{\phi}_6 = \ddot{\theta}_{c2} - \ddot{\theta}_t \quad (34)$$

Now, the ith control constraint, could be composed into a second order differential equation as

$$\ddot{\phi}_i + \alpha_i \dot{\phi}_i + \gamma_i \phi_i = 0 \quad (35)$$

where the constants α_i, $\gamma_i > 0$ determine the performance of ith the control action. Substituting Eqs. (16)-(24) and Eqs. (26)-(34) into Eq. (35) and rearranging to keep controlled accelerations to the left hand side,

$$\begin{aligned}
\ddot{x}_{c1} &= \ddot{x}_t - L_0 \sin\theta_t \ddot{\theta}_t - L_0 \cos\theta_t \dot{\theta}_t^2 - \alpha_1(\dot{x}_{c1} - \dot{x}_t \\
&\quad + L_0 \sin\theta_t \dot{\theta}_t) - \gamma_1(x_{c1} - x_t - L_0 \cos\theta_t) \\
&= b_1 \quad (36) \\
\ddot{y}_{c1} &= \ddot{y}_t + L_0 \cos\theta_t \ddot{\theta}_t - L_0 \sin\theta_t \dot{\theta}_t^2 - \alpha_2(\dot{y}_{c1} - \dot{y}_t \\
&\quad - L_0 \cos\theta_t \dot{\theta}_t) - \gamma_2(y_{c1} - y_t - L_0 \sin\theta_t) \\
&= b_2 \quad (37) \\
\ddot{\theta}_{c1} &= \ddot{\theta}_t - \alpha_3(\dot{\theta}_{c1} - \dot{\theta}_t) - \gamma_3(\theta_{c1} - \theta_t - \pi) \\
&= b_3 \quad (38) \\
\ddot{x}_{c2} &= \ddot{x}_t + L_0 \sin\theta_t \ddot{\theta}_t + L_0 \cos\theta_t \dot{\theta}_t^2 - \alpha_4(\dot{x}_{c2} - \dot{x}_t \\
&\quad - L_0 \sin\theta_t \dot{\theta}_t) - \gamma_4(x_{c2} - x_t + L_0 \cos\theta_t) \\
&= b_4 \quad (39) \\
\ddot{y}_{c2} &= \ddot{y}_t - L_0 \cos\theta_t \ddot{\theta}_t + L_0 \sin\theta_t \dot{\theta}_t^2 - \alpha_5(\dot{y}_{c2} - \dot{y}_t \\
&\quad + L_0 \cos\theta_t \dot{\theta}_t) - \gamma_5(y_{c2} - y_t + L_0 \sin\theta_t) \\
&= b_5 \quad (40) \\
\ddot{\theta}_{c2} &= \ddot{\theta}_t - \alpha_6(\dot{\theta}_{c2} - \dot{\theta}_t) - \gamma_6(\theta_{c2} - \theta_t) \\
&= b_6 \quad (41)
\end{aligned}$$

Note that, since the target is passive, the control action takes place in the chaser alone. Hence, the target states are part of the right hand side b_i for $i = 1, 2 \ldots 6$ in Eqs. (36)-(41).

In matrix form, Eqs. (36)-(41) are written the standard control constraint form as

$$\mathbf{A}_c \ddot{\mathbf{q}} = \mathbf{b}_c \quad (42)$$

where

$$\begin{aligned}
\mathbf{A}_c &= \mathbf{I}_{6\times 6} \quad (43) \\
\ddot{\mathbf{q}} &= \begin{bmatrix} \ddot{x}_{c1} & \ddot{y}_{c1} & \ddot{\theta}_{c1} & \ddot{x}_{c2} & \ddot{y}_{c2} & \ddot{\theta}_{c2} \end{bmatrix}^\intercal \\
\mathbf{b}_c &= \begin{bmatrix} b_1 & b_2 & b_3 & b_4 & b_5 & b_6 \end{bmatrix}^\intercal
\end{aligned}$$

Using Eq. (42) in Eq. (14),

$$\mathbf{F}_c = \mathbf{A}_c^\intercal (\mathbf{A}_c \mathbf{M}^{-1} \mathbf{A}_c^\intercal)^+ (\mathbf{b}_c - \mathbf{A}_c \mathbf{M}^{-1} \mathbf{F}_u) \quad (44)$$

where $\mathbf{M} = \text{diag}(\begin{bmatrix} m_{c1} & m_{c1} & J_{c1} & m_{c2} & m_{c2} & J_{c2} \end{bmatrix})$ is the generalized mass matrix. In actual implementation, the generalized force command has to be sent to a control mixer block that can convert the commands to appropriate thruster firings.

V. Simulations

Simulations are performed to demonstrate the proposed close-range rendezvous. The chasers and target are assumed to have constant mass of $m_k = 1$ kg and inertia of bodies along the out-of-plane axis is taken as $J_k = 0.5$ kg$\cdot$m^2. The desired stand-off separation between the target and chaser is $L_0 = 10$ m. For the ith control constraint, the gains in Eq. (35) are chosen as $\alpha_i = 2$ and $\gamma_i = 1$. Note that in the real implementation, the gains would have to be adjusted so as to fit the performance requirement. External forces are ignored on account of the high magnitude of the chaser thrust and a fast rendezvous is assumed to approximate a double integrator dynamics of the kth body as

$$\begin{bmatrix} \ddot{x}_k \\ \ddot{y}_k \\ \ddot{\theta}_k \end{bmatrix} = \begin{bmatrix} m_k & 0 & 0 \\ 0 & m_k & \\ 0 & 0 & J_k \end{bmatrix}^{-1} \begin{bmatrix} F_{xk} \\ F_{yk} \\ \tau_k \end{bmatrix} \quad (45)$$

where F_{xk} and F_{yk} are the x and y-components of the inertial force applied on the kth body and τ_k is the moment applied on the body frame.

Case 1: Single chaser rendezvous

Chaser 1 is initially located at $(5, 2.5)$ m at an orientation π rad. Corresponding linear velocity is $(0.5, 0.5)$ m/s while the out-of plane initial angular velocity is 0.9 rad/s. The target initially located at the origin has a constant velocity of $(-1, 0.3)$ m/s. It is also spinning at a constant rate of 1.2 rad/s, starting from an initial orientation of 0 rad. Figure. 4a presents the planar rendezvous of a single chaser around a target. It can be seen that the method allows to align a vector on the chaser frame at a specific angle with respect to a vector on target, as observed from the inertial frame. The time history of the position of both the uncontrolled target and the controlled chaser is plotted in Fig. 4b. Fig. 4c presents the plot of the angular displacements of the target and Chaser 1. The control forces and torques acting on the chaser as shown in Fig. 4d ensures that the corresponding relative separation also settles to the desired stand-off distance of $L_0 = 10$ m as observed from Fig. 4f, while simultaneously matching their angular velocities as in Fig. 4e.

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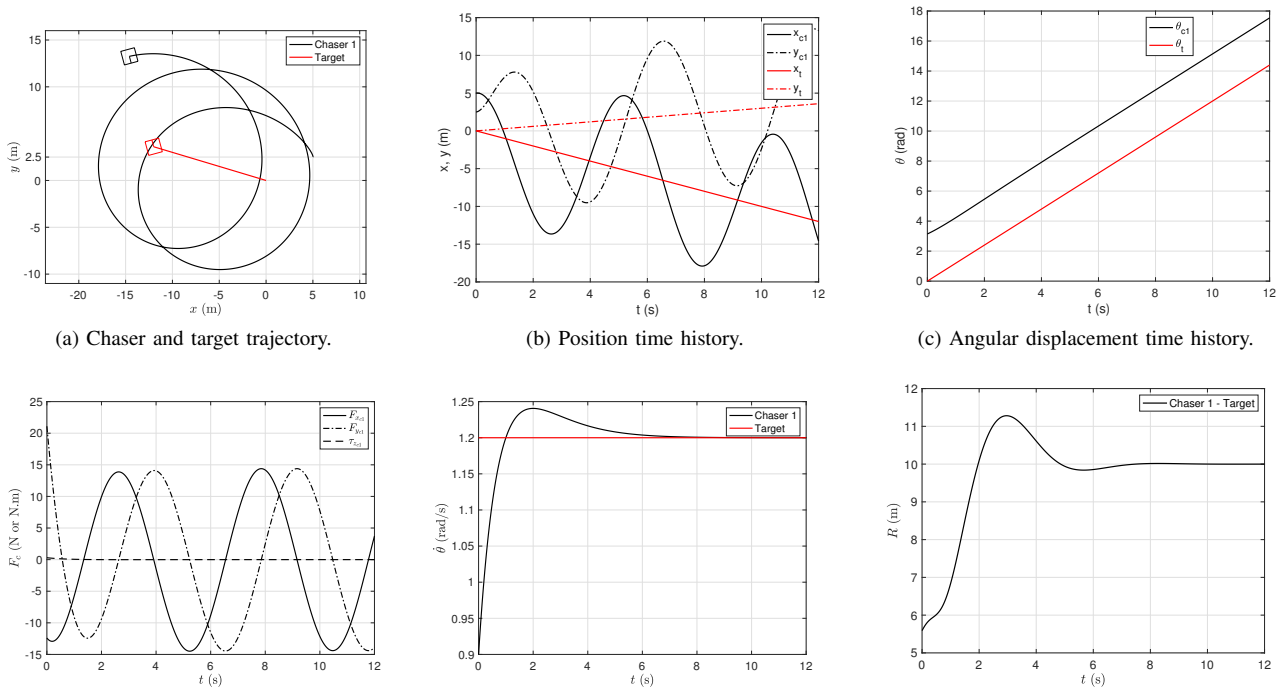
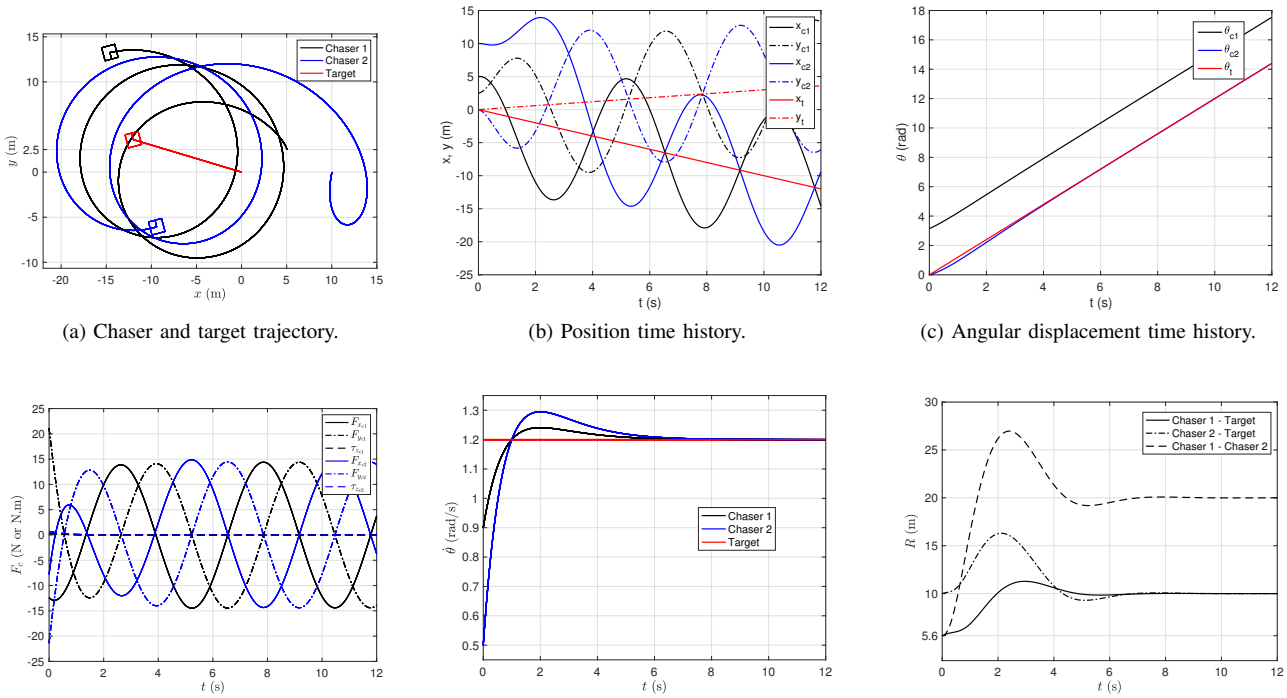


(a) Chaser and target trajectory. (b) Position time history. (c) Angular displacement time history.

(d) Control force. (e) Absolute angular velocity. (f) Stand-off separation.

Fig. 4: Results for Case 1: Single-chaser rendezvous.



(a) Chaser and target trajectory. (b) Position time history. (c) Angular displacement time history.

(d) Control force. (e) Absolute angular velocity. (f) Stand-off separation.

Fig. 5: Results for Case 2: Dual-chaser rendezvous.

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Case 2: Dual-chaser rendezvous

In this scenario, the initial states for both the Chaser 1 and target are assumed to be the same as in Case 1. Further, Chaser 2 is initially positioned at $(10, 0)$ m with an alignment of 0 rad with respect to inertial x-axis, while the initial linear and angular velocities are $(0.1, -1)$ m/s and 0.5 rad/s, respectively. Fig. 5a presents the dual-chaser rendezvous of the spinning target. Note that, the vector on Chaser 2 frame aligns along the target vector in the inertial frame. This configuration can, potentially, be useful in aligning multiple chasers prior to deploying tethers that can stabilize the target. The corresponding evolution of the position and angular displacements are plotted in Fig. 5b and Fig. 5c, respectively. Control forces and torques shown in Fig. 5d ensure that the constraints of Eqs. (16)-(28) are met as governed by Eq. (44). The simultaneous angular velocity synchronization and stand-off distance maintenance can be seen from Fig. 5e and Fig. 5f, respectively.

VI. Conclusion

Udwadia-Kalaba framework is proposed for simultaneous position and angular orientation control of a single chaser and a dual chaser system in close-range rendezvous with respect to a spinning target. Simulations are carried out that demonstrate the asymptotic convergence onto the stand-off distance with respect to the target, while aligning with it. The formulation is ideal for performing a validation of the UKE-based control technique on a planar proximity operations testbed.

The future work would focus on the experimental validation of this work. Further, a spatial control would be developed and the effect of parameter uncertainties, noise and orbital perturbations on the control performance would be studied. Another direction of interest include the determination of minimum-fuel solutions in the spatial case.

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